Why crop rankings do not determine land allocation

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Abstract

Crop rankings are ordinal, whereas acreage allocation is a cardinal decision made under joint uncertainty and operational constraints. We formalize this gap with a set-valued, multi-crop programme and distinguish possible, universal and selected ranking reversal. Exact results show that rankings alone do not identify acreage and that operational feasibility can force universal reversal even when downside risk is inactive. In pre-specified simulations, assigned operational interventions convert a corn-only allocation into soybean-majority allocations in every treated cell; all 24/24 family-wise intervals meet the precision criterion. A separate information–flexibility experiment shows that their interaction can be positive (3.258, 95% interval 2.788 to 3.728), exactly null or substitutive. Four other pre-specified experiments do not meet their experiment-level precision criteria and remain inconclusive. An official US panel extending from 2016 to 2024 contains 248 complete state-years in 31 states. Concurrent operating-margin inversion intensity is 0.411 (state-cluster 95% interval 0.346 to 0.475), but all four strictly lagged acreage-share intervals include zero and national summaries depend on aggregation. Ranking reversal is therefore demonstrable under explicit mechanisms, while public acreage disagreement alone does not identify private objectives, constraints, risk preferences, dependence, causality or welfare.

1 Introduction

Crop prediction systems increasingly turn weather, soil, management and remote-sensing data into yield forecasts, suitability scores and crop rankings [Liakos et al., 2018, van Klompenburg et al., 2020]. These outputs answer an ordinal question—which crop scores higher—but not the operational question of how much acreage to assign to each crop. A ranking remains unchanged under every strictly increasing transformation of its scores; an acreage allocation generally does not. Moving from prediction to prescription therefore requires cardinal margins, a joint uncertainty model, operational constraints and a rule for selecting among multiple optima.

Crop-planning studies represent acreage as a constrained portfolio decision with resource limits, rotations, stochastic returns, robustness or downside risk [Filippi et al., 2017, Boyabatlı et al., 2019, Benini et al., 2023, Randall et al., 2022, 2024]. Prescriptive analytics and decision-focused learning likewise separate predictive accuracy from downstream decision quality [Bertsimas and Kallus, 2020, Elmachtoub and Grigas, 2022, Wilder et al., 2019, Mandi et al., 2023]. Yet a score–acreage mismatch alone does not reveal whether a decision is constrained, risk-sensitive, inefficient or even unique. The same visual pattern can arise from different mathematical mechanisms.

1.1 Position relative to crop planning and prescriptive learning

Agricultural planning models optimize a cardinal decision. Their variables are planted areas, crop sequences or resource assignments; their objectives are profit, cost, environmental performance or a declared multi-objective criterion; and their restrictions encode land, labour, capital, rotations and agronomic requirements [Filippi et al., 2017, Boyabatlı et al., 2019, Benini et al., 2023]. This literature establishes that crop choice is operationally coupled. It does not imply that an externally produced suitability order contains the cardinal objective coefficients or recovers the feasible set. Our question begins one step earlier: what can be inferred about an acreage vector when the public object is only a crop order?

Risk-aware crop planning adds scenarios, robust uncertainty sets or downside-risk restrictions [Randall et al., 2022, 2024]. Loss-CVaR is useful because it represents a tail expectation and admits a finite-scenario linear formulation [Rockafellar and Uryasev, 2000, 2002]. Dependence matters because crop prices and yields enter portfolio profit jointly; copula families provide one way to hold marginals fixed while varying that joint law [Demarta and McNeil, 2005, Ansari and Rockel, 2024]. These tools specify a decision problem, but they do not make tail dependence, a binding risk limit or diversification identifiable from a rank–acreage mismatch. We therefore treat dependence and downside risk as candidate mechanisms with explicit assumptions, not as labels assigned to observed state-years.

Prediction-to-prescription and decision-focused learning optimize predictive systems for downstream objectives rather than predictive loss alone [Bertsimas and Kallus, 2020, Elmachtoub and Grigas, 2022, Wilder et al., 2019, Mandi et al., 2023]. That distinction is closely related to ours but concerns a different inferential object. A decision-focused learner is evaluated against a declared optimization problem; a published ranking does not disclose one. Likewise, rank-based crop recommendation is attractive because ranks are comparable and easy to communicate, yet the invariance that makes a rank robust also removes information about payoff gaps. We formalize the missing bridge rather than assuming a proportional score-to-acreage rule.

Finally, the value of a signal depends on the decisions it can change. Classical value-of-information analysis compares optimized values with and without information, while flexibility changes the available action set [Keisler, 2004, Merkhofer, 1975, Van Mieghem, 1998]. Neither value is defined by ranking accuracy alone. Our information experiment uses constructive finite-state environments to show that information value is non-negative when a signal can be ignored, but its interaction with a larger action set can be positive, zero or negative. The paper consequently links crop planning and prescriptive learning through a common identification boundary: the decision model must be stated before a ranking can carry operational meaning.

Here we reconstruct the ranking-to-allocation argument as a closed evidence chain (Fig. 1). A set-valued stochastic programme maximizes expected margin over a polyhedral operational set under loss-CVaR, using the atom-safe representation of Rockafellar and Uryasev [2000, 2002]. Reversal can occur at some optimum, every optimum or only the selected optimum. Exact constructions separate cardinal-margin, operational, downside-risk and multiplicity mechanisms, while a complete Karush–Kuhn–Tucker identity records local pressure without converting marginal conditions into acreage ranks.

Six pre-specified experiments test this mechanism framework. Only the assigned operational-intervention and information–flexibility experiments meet every experiment-level precision criterion. The former produces universal reversal throughout the treated cells. The latter yields positive, exact-null and substitutive interactions across three constructive environments. Four other experiments reach their replication ceilings without meeting every pre-specified interval criterion; their complete results remain visible in the Supplementary Information but do not support positive claims.

The empirical analysis asks a narrower question using official observations. The reconstructed panel covers 2016–2024, 248 complete state-years and 31 states. Concurrent score–acreage inversion is definition- and state-dependent, whereas all four strictly lagged acreage-share intervals include zero and national aggregation changes the pattern. These observations document descriptive disagreement, not farm objectives or optimal acreage. The contribution is consequently both positive and bounded: explicit mechanisms can generate ranking reversal, but public rank–acreage disagreement cannot select among those mechanisms without decision-level information.

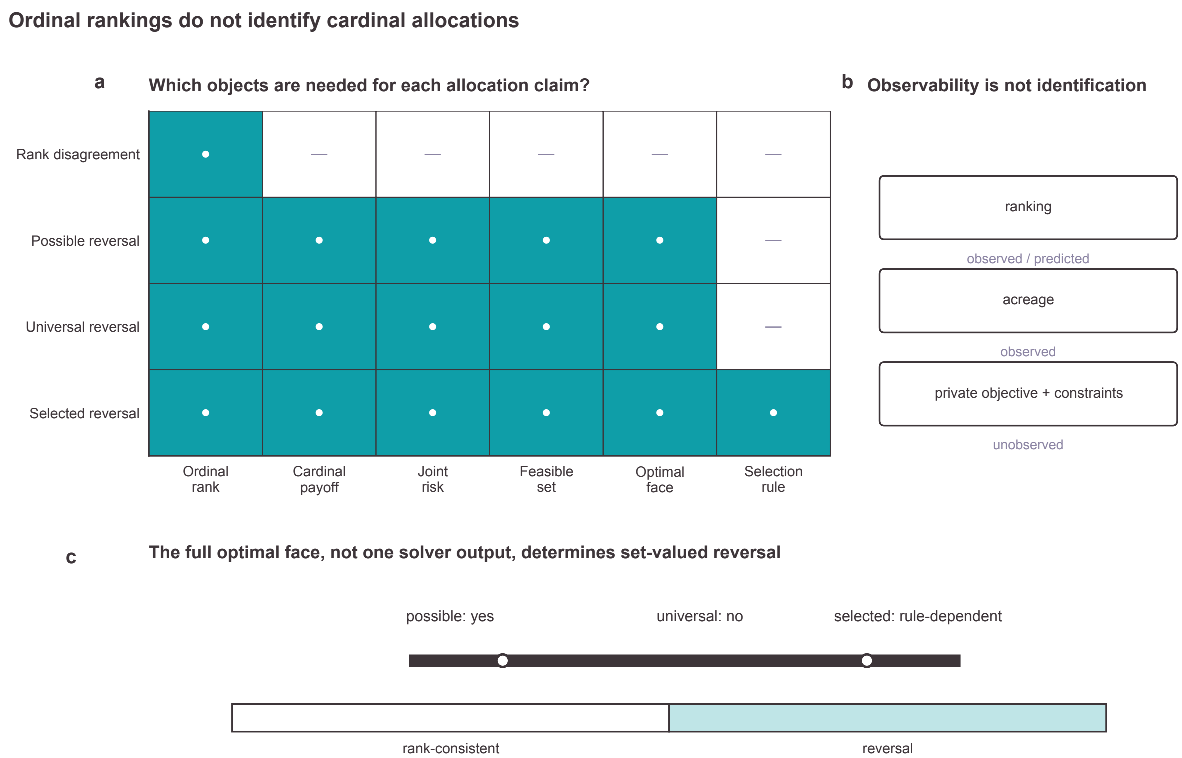

Figure 1: From crop ranking to an allocation claim. Cardinal payoffs, joint uncertainty, an operational feasible set and a selection rule intervene between an ordinal ranking and an acreage vector. Possible and universal reversal are properties of the full optimal face; selected reversal depends on a declared selection rule.

2 Ranking does not identify allocation

A crop ranking and an acreage allocation are different mathematical objects. Let a suitability vector s induce a weak order and let $x \in \mathbb{R}^n_+$ denote acreage. Strictly increasing transformations preserve the order in s but can change any proportional score-to-acreage rule. The ranking also supplies neither an objective nor a feasible set. Consequently, even a strict order cannot identify a cardinal acreage vector.

2.1 Stochastic profit, feasibility and tail risk

For crop i and scenario ω, write per-acre stochastic margin and portfolio profit as

$$\widetilde{\pi}_i(\omega) = \widetilde{P}_i(\omega)\widetilde{Y}_i(\omega) - C_i, \qquad \widetilde{\Pi}(x,\omega) = \sum_{i=1}^{n} x_i \widetilde{\pi}_i(\omega). \tag{1}$$

Price and yield may be dependent within and across crops. The mean-margin vector $\mu = \mathbb{E}[\widetilde{\pi}]$ supplies cardinal levels and gaps that an ordinal score does not. The joint law of $\widetilde{\pi}$, not its crop-wise marginals alone, determines portfolio loss $L(x,\omega) = -\widetilde{\Pi}(x,\omega)$ and its tail.

The operational set records crop bounds, total land, budget, rotations, contracts and other shared linear requirements. With lower and upper bounds ℓ, u, per-acre cash requirement c, land A, budget B and shared restrictions $Gx \le g$, it is

$$X = \{x \in \mathbb{R}^n_+ : \ell \le x \le u,\ \mathbf{1}^\top x \le A,\ c^\top x \le B,\ Gx \le g\}. \tag{2}$$

Table 1: Objects required for ranking-to-allocation inference. Public observability does not by itself establish that observed acreage solves the canonical programme.

Object	Typical status	Role in the decision	Claim available without it
Suitability score s	Observed or predicted	Supplies an ordinal weak order	Rank agreement only
Margin vector μ	Calibrated or private	Gives cardinal payoff levels and gaps	No value or acreage comparison
Joint law of $\widetilde{\pi}$	Estimated or designed	Determines portfolio and tail loss	No dependence or risk mechanism
Operational set X	Partly private	Defines land, budget, rotation and contract choices	No feasibility attribution
Optimal face S_κ	Model-derived	Contains every objective-equivalent optimum	No possible/universal classification
Selector σ	Declared decision rule	Chooses one reported optimizer	No selected-reversal claim

Rotation and contract restrictions can enter $Gx \leq g$ or tighten crop-specific bounds. This polyhedral representation makes feasibility and active faces transparent while allowing the risk-constrained problem to remain a linear programme under finite scenarios.

For confidence level α, the canonical programme maximizes expected margin subject to a loss-CVaR ceiling κ,

$$\max_{x \in X} \mu^\top x \quad \text{subject to} \quad \mathrm{CVaR}_\alpha[L(x)] \leq \kappa. \tag{3}$$

For scenarios $k = 1, \ldots, m$ with probabilities w_k, a free threshold v and excess variables $q_k \geq 0$ give the linear representation

$$v + \frac{1}{1-\alpha} \sum_{k=1}^{m} w_k q_k \leq \kappa, \qquad q_k \geq -\pi_k^\top x - v. \tag{4}$$

Equations 3–4 separate expected value, downside risk and operations. They also show why a score order cannot reveal which part of the decision system is active.

The distinction becomes sharper when the optimum is set-valued. Let $r_\alpha(x) = \mathrm{CVaR}_\alpha[-\widetilde{\pi}^\top x]$ be loss-CVaR. The optimal face is

$$S_\kappa = \arg\max\{\mu^\top x : x \in X,\ r_\alpha(x) \leq \kappa\}. \tag{5}$$

For a ranked pair $s_i > s_j$, reversal is *possible* if $x_j > x_i$ at some $x \in S_\kappa$, *universal* if it holds at every $x \in S_\kappa$, and *selected* if it holds at the allocation returned by a declared deterministic rule. These definitions prevent one solver vertex from being mistaken for the entire optimal face.

The selected comparison uses one deterministic selector $\sigma(S_\kappa)$ held fixed across counterfactuals. This makes replay reproducible; it does not convert selection-dependent reversal into a property of every optimizer. Computational classification therefore solves two additional linear programmes on the objective-equivalent face: one minimizes and one maximizes $x_j - x_i$. Their signs supply universal and possible reversal independently of the reported vertex.

Four exact two-crop constructions share the same coordinate system (Fig. 2). Cardinal margins can point against the external crop ranking. A bound or contract can place the entire feasible set in

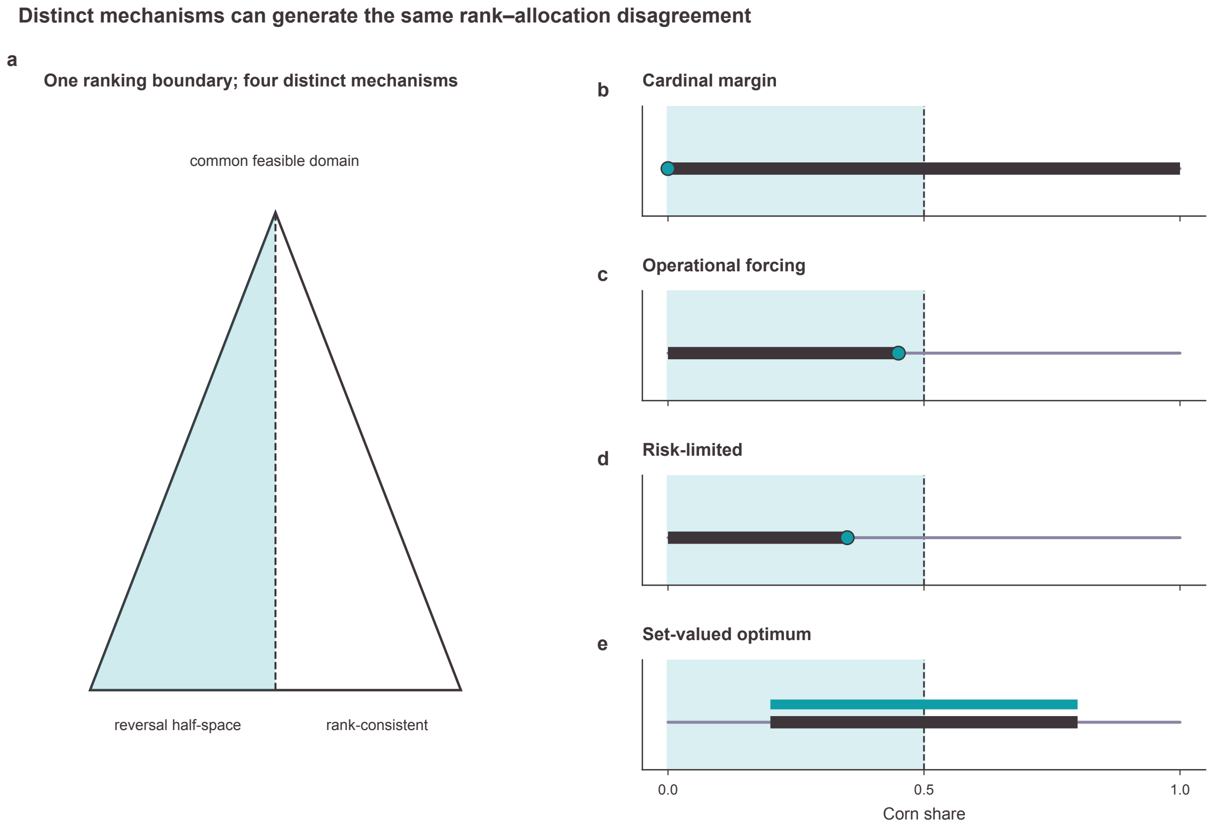

Figure 2: A four-mechanism atlas on shared coordinates. Cardinal-margin, operational, downside-risk and set-valued cases use the same corn-share axis and reversal boundary. The cases are exact analytic constructions, not empirical estimates.

the reversal half-space. A binding downside-risk constraint can move the optimum without implying monotone crop-wise comparative statics. A flat optimal face can cross the rank boundary, making reversal possible but not universal. Identical-looking allocations can thus encode different claims.

Operational feasibility supplies a transparent universal certificate. If $s_i > s_j$ and

$$\sup_{x \in X} x_i + \tau_x < \inf_{x \in X} x_j, \tag{6}$$

then every feasible allocation reverses the pair; $u_i + \tau_x < \ell_j$ is a simple sufficient condition. This result needs no downside-risk channel. Conversely, if an unconstrained expected-margin optimum is risk-feasible, the risk constraint leaves the optimal value unchanged, but one slack selected solution does not establish equality of optimal faces when optima are multiple.

The certificate is positive rather than merely cautionary. Once a lower Soybean requirement or Corn upper bound places all feasible allocations beyond the rank boundary, universal reversal follows before the objective or tail-risk term is evaluated. The geometric construction connects directly to the assigned operational experiment below. By contrast, the risk-limited construction is an exact existence result: it shows that downside risk can expose a reversing face, not that increasing dependence or tightening a CVaR ceiling must move every crop monotonically.

The complete first-order identity blocks a second shortcut. At an optimum, non-negative multipliers and a subgradient $d \in \partial r_\alpha(x)$ satisfy

$$0 = -\mu + \lambda d + \gamma \mathbf{1} + \beta c + G^\top \eta - a + b. \tag{7}$$

For crops i and j,

$$\mu_i - \mu_j = \lambda(d_i - d_j) + \beta(c_i - c_j) + (G^\top \eta)_i - (G^\top \eta)_j - (a_i - a_j) + (b_i - b_j). \tag{8}$$

The common land multiplier cancels, whereas budget, shared-constraint and bound terms remain. Equation 8 describes local marginal balance; it does not rank acreage levels.

The identity nevertheless identifies pressure channels inside a declared model. The first term is the local loss-CVaR subgradient difference, the second is crop-specific budget pressure, the third collects shared operational restrictions, and the last two are lower- and upper-bound pressures. A non-zero term explains how stationarity is balanced at the solution. It is neither a global decomposition of planted acres nor evidence that the corresponding private restriction exists in an observational state-year.

The closed M0–M4 path makes these distinctions auditable (Fig. 3). Successive stages add cardinal margins, operations, downside risk and dependence to a common benchmark. Allocation and indexed outcome panels retain the actual registered outputs; an all-subset Shapley decomposition reports attribution intervals and closes with maximum efficiency residual 1.42×10^{-14}. Signed E2 pressure terms show how margin pressure is balanced locally by budget, shared or boundary terms, with maximum stationarity residual 1.82×10^{-11}. These are accounting decompositions over a controlled model, not causal effects in observed acreage.

3 Operational constraints identify reversal

E2 assigns budget, rotation, contract and dominant-crop-bound interventions while holding the ranking and scenario law fixed (Fig. 4). The base allocation assigns corn share 1.000. Each treated cell instead assigns a soybean majority: contract or rotation interventions produce corn and soybean shares 0.450 and 0.550, while the tighter budget produces 0.318 and 0.682. Possible, universal and selected reversal coincide in every treated cell, including the bound anchor.

The intervention factorial distinguishes three claims. First, the factor matrix records which components of the feasible set are assigned rather than inferred. Second, optimization over the complete objective-equivalent face asks whether reversal is possible, universal or dependent on the selected vertex. Third, the contrast family measures allocation displacement, expected-margin change and the change in selected reversal probability against the common baseline. The risk ceiling and scenario law are identical across cells, so changes cannot be attributed to a redesigned dependence model.

All 24 of 24 registered family-wise intervals meet their precision targets at 16 replications per cell. The optimal-face classifications rule out selection dependence, and the local KKT audit shows that the loss-CVaR term is inactive in these cells. Because the intervention is assigned within a closed simulation, E2 identifies an operational displacement mechanism in its declared domain. It does not estimate how often that mechanism operates in agriculture.

The mechanism fingerprint clarifies the allocation response. In the baseline, margin pressure is balanced at the Corn boundary. Contract or rotation changes expose a Soybean-majority solution at Corn share 0.450, while the budget intervention moves the solution farther to 0.318. Direct-forcing cells combine restrictions whose KKT pressures jointly balance the same margin difference. The zero tail-risk pressure rules out downside risk as the active channel in this experiment; it does not imply that risk is generally irrelevant to crop allocation.

Inference is experiment-level rather than contrast-level. E1, E3, E4 and E5 each reach their pre-specified ceiling without meeting every primary precision criterion. Isolated precise contrasts therefore do not support experiment-level conclusions. E1 is reported in Supplementary Fig. S1, E3 in Supplementary Fig. S2, and E4/E5 in Supplementary Fig. S3. All 206 infeasible outcomes remain in the complete results; no feasible-only subset enters a positive claim.

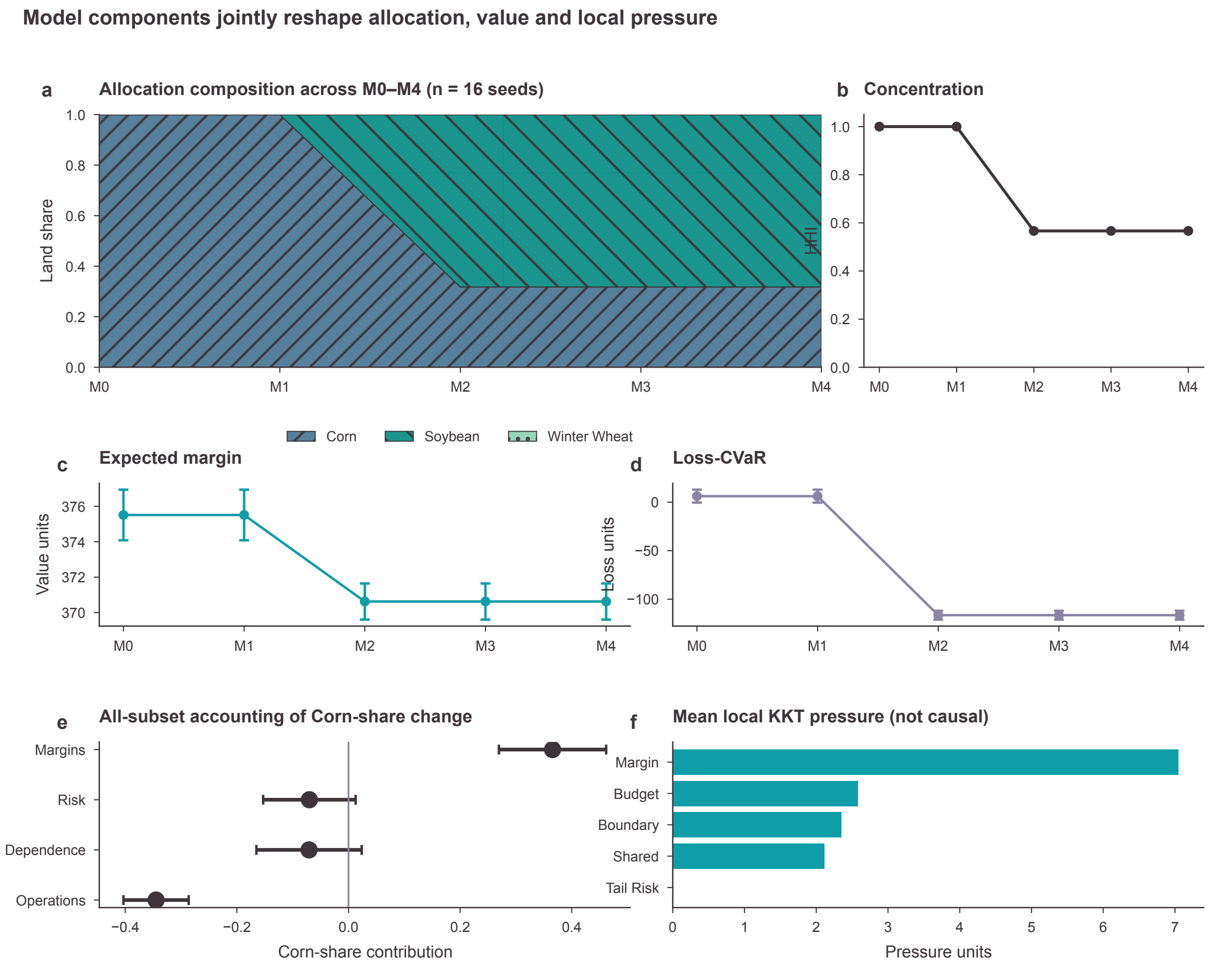

Figure 3: Nested allocation and mechanism decomposition. The M0–M4 path reports crop allocation, indexed outcomes, all-subset Shapley contributions and signed E2 KKT pressures. Intervals summarize registered closed-domain repetitions; pressure terms are local accounting identities rather than causal acreage effects.

4 Information value depends on available actions

E6 crosses signal information with action-set flexibility in three constructive environments. Define the interaction as the difference-in-differences between informative and uninformative signals under flexible and restricted action sets. Ignoring a signal remains feasible, so information value is non-negative in each action set; exact ignore-signal anchors and signal-garbling checks verify this implementation invariant.

Decision timing is explicit. Nature first draws a state, the decision-maker observes a signal of registered accuracy and then chooses an action from either the restricted or flexible set. The no-information benchmark chooses one action before observing the signal. Because a contingent policy may ignore the signal and repeat that action, information cannot reduce optimized value. Flexibility separately enlarges the feasible action set. The interaction compares how much the signal gains from that enlargement; it is not the level of information value itself.

The interaction spans all three theoretically admissible cases (Fig. 5). When flexibility unlocks state-contingent specialization, the estimate is 3.258 with 95% family-wise interval 2.788 to 3.728. A robust fallback makes information and flexibility substitutes, giving -3.114 with interval -3.476 to

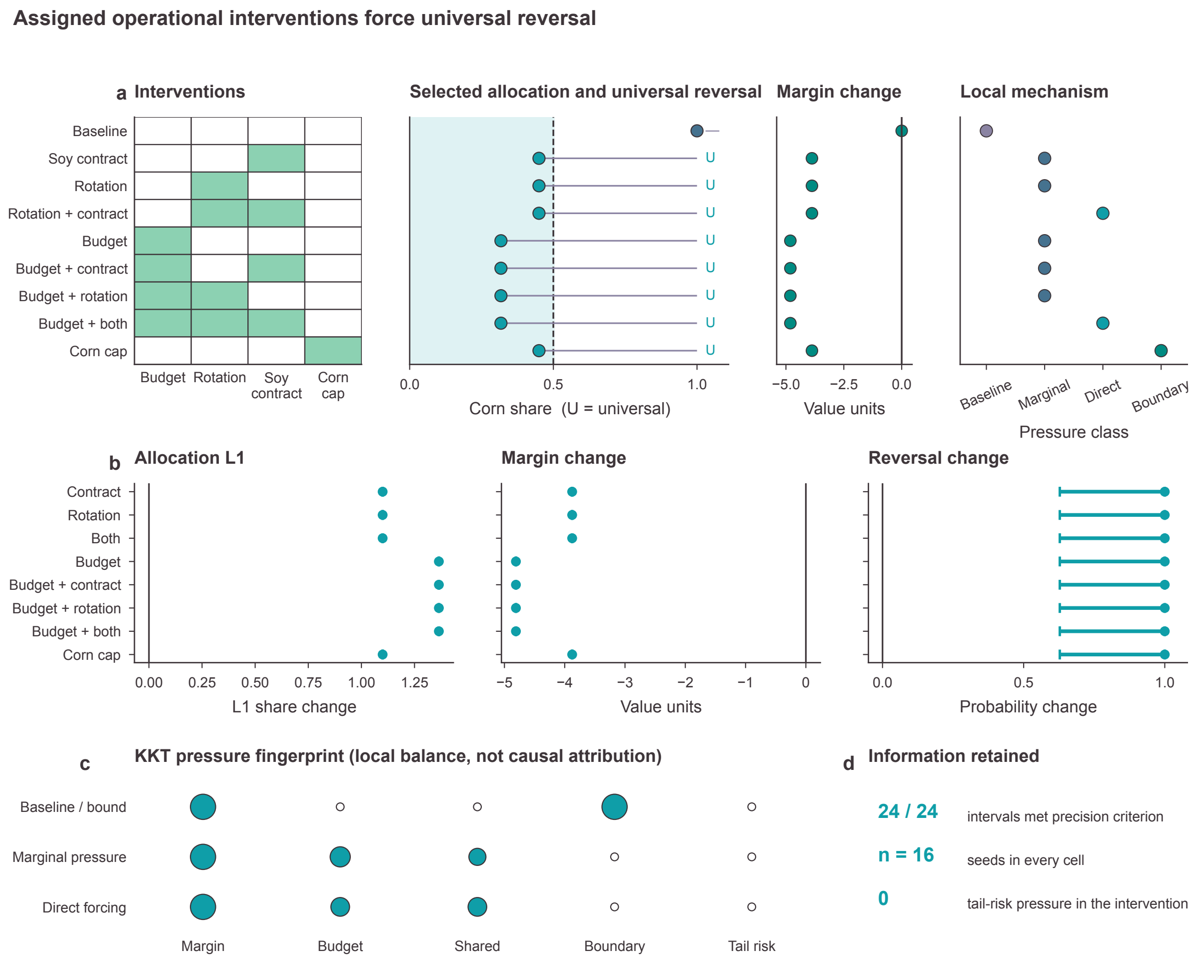

Figure 4: Assigned operational interventions identify reversal. The display reports the factorial allocation, selected and optimal-face classifications, family-wise contrasts and mechanism-class summaries. Each cell has $n = 16$ and all $24/24$ intervals meet the pre-specified precision criterion.

-2.751. A dominated added option yields the exact-null interaction 0.000. All three intervals meet the precision criterion at 16 replications.

This sign pattern shows why information and flexibility are not universally complementary. Information is actionable when it changes a feasible contingent decision, but the incremental value of the signal depends on what the enlarged action set adds. E5 cross-law dependence diagnostics are therefore kept separate in Supplementary Fig. S3: its experiment-level precision criterion was not met, so it cannot support a general claim about copula misspecification, regret or downside risk.

The three archetypes supply constructive intuition. Under specialization unlocks, the flexible set adds state-contingent actions that make a more accurate signal useful. Under the dominated option, the added action is never optimal, leaving both the flexibility gain and the interaction at zero. Under robust-option substitutes, flexibility adds an action that performs well without resolving the state, reducing the incremental value of information. These examples establish the absence of a universal sign; they are not estimates of the frequency or magnitude of complementarity in agricultural decisions.

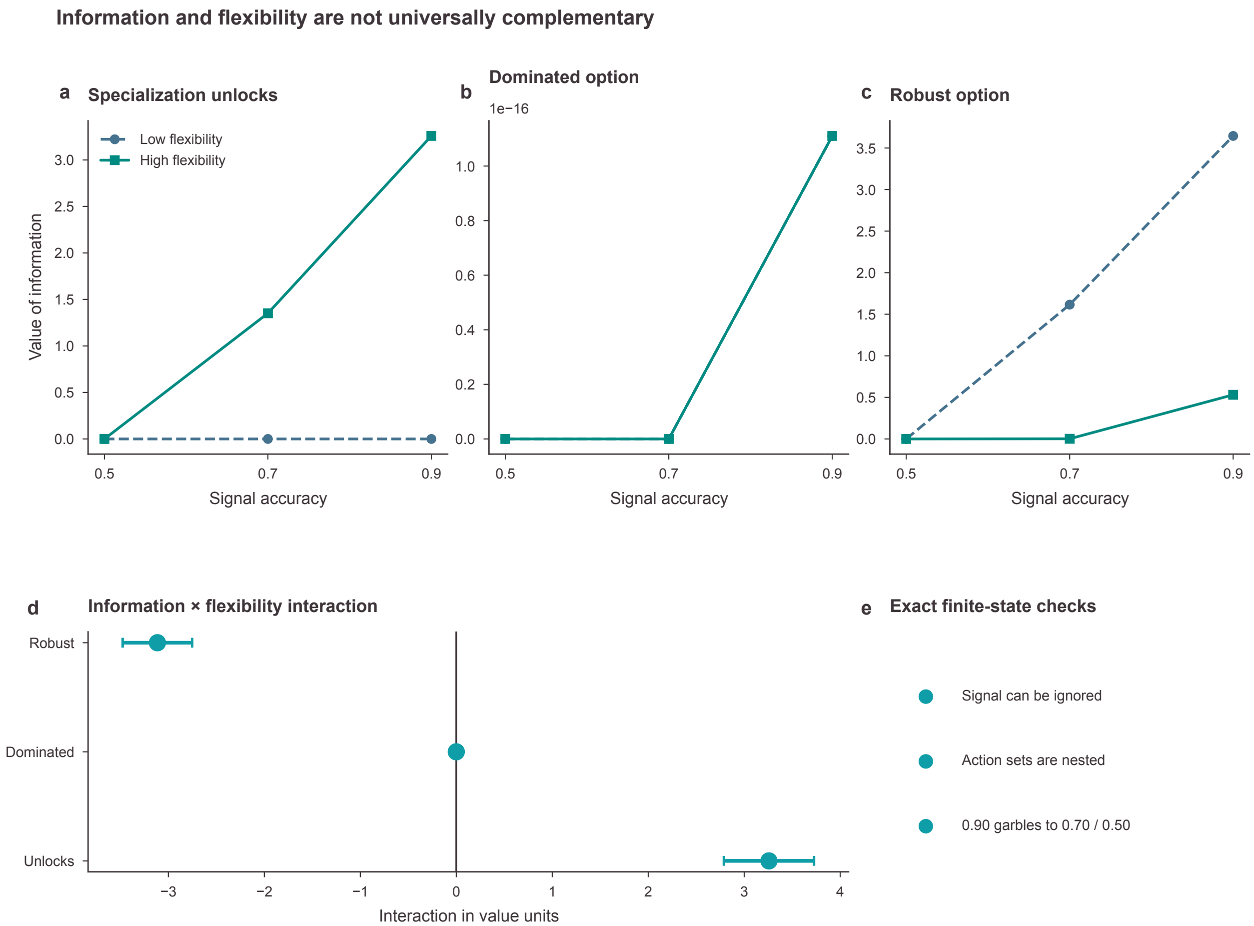

Figure 5: Information and action-set flexibility can complement, substitute or be unrelated. The display reports three pre-specified archetypes, their interaction intervals, exact finite-state checks and archetype-specific flexibility gains. All three family-wise intervals meet the pre-specified precision criterion.

5 Empirical patterns are definition-dependent

The official-data reconstruction extends the common state panel to 2016–2024 (Fig. 6). It contains 744 complete state–crop–year rows, 248 complete state-years and 31 states. State acreage and yield come from non-overlapping USDA National Agricultural Statistics Service annual-report blocks. National prices and costs come from USDA Economic Research Service series, and the registered CPI-U vintage converts monetary variables to real dollars. Missing or suppressed acreage and yield are not imputed.

5.1 Sample construction and score definitions

The parser reads 1035 state–crop–year rows. Requiring both acreage and yield retains 971 rows; requiring all three crops in the same state-year retains 744 crop rows, or 248 state-years. Coverage is 31 states in each of the first three years, 26 states from 2019 through 2023 and 25 states in 2024. The source audit records 1 missing acreage value and 64 missing yield values. Complete-case selection is therefore visible rather than treated as an implicit balanced panel.

Four accounting definitions deliberately vary what is meant by a "high" crop score. Relative yield compares state yield with the same crop's national yield. Standardized revenue multiplies state yield

Table 2: Empirical score definitions and interpretation. Every definition is constructed before comparison with observed planted-acreage shares.

Definition	Construction	Interpretation boundary
Relative yield	State yield divided by same-year national yield for that crop	Productivity index; nationally tied by construction
Standardized revenue	State yield times national crop price, standardized within state-year	Gross accounting score, not local revenue
Operating margin	Standardized revenue less national operating cost	Excludes local basis and heterogeneous cost
Total-cost margin	Standardized revenue less national total cost	Broad accounting score, not private expected profit

by the national crop price. Operating margin subtracts operating cost, whereas total-cost margin subtracts the broader cost measure. Prices and costs are national inputs, so neither margin is a farm-level realized profit. Acreage share is always reported as a proportion of the three-crop total in estimation and as percentage-point change when multiplied by 100 in figures.

Concurrent score and acreage orders differ materially. Operating-margin inversion intensity is 0.411 with a state-cluster 95% interval of 0.346 to 0.475, and its score leader differs from the acreage leader in 62.5% of complete state-years. Relative yield, standardized revenue and total-cost margin give different patterns, showing that an inversion statement inherits the accounting definition used to construct the score. The ranked state display further shows that the operating-margin pattern is not spatially uniform.

Inversion intensity counts discordant crop pairs among the three available pairs and therefore ranges from zero to one. Top-rank disagreement asks only whether the score and acreage leaders differ. Kendall's tau-b retains ties and supplies an ordinal association measure even when two crops receive the same score; Spearman correlation is reported when defined. These estimands answer different questions and are not interchangeable. State distributions and annual paths in Fig. 6 show that definition sensitivity is present both across place and over time rather than being an artefact of one national mean.

The strictly lagged analysis does not provide positive evidence that a previously leading score predicts a larger subsequent acreage-share change. For operating margin, the prior-leader coefficient is 0.0021 with interval -0.0041 to 0.0084. The other three primary intervals also include zero. Across the 217 transitions per definition, the acreage leader changes in 24 operating-margin transitions while score-leader movement is definition-specific. This is compatible with persistence, but the aggregate data do not measure rotations, adjustment costs, contracts or farmer expectations.

The temporal specification compares next-year acreage-share change for the prior score leader with other crops while controlling for prior acreage share and crop, decision-year and state fixed effects. It contains 2604 crop-transition rows, uses 5000 state-cluster draws and never uses a contemporaneous score to predict the same year's share. A rank-sensitivity specification replaces the top-score indicator with prior rank. Persistence and transition summaries separately classify whether the score leader, acreage leader, both or neither change between adjacent years. These summaries describe adjustment patterns; they do not supply an instrument or a causal adoption design.

Aggregation changes the empirical statement. Operating-margin inversion intensity is 0.148 nationally and total-cost-margin inversion is 0.481. Relative yield is tied by construction in all 9 national years and is therefore non-informative rather than rank-aligned. The national summaries neither validate a preferred score nor negate state heterogeneity; they show that the spatial support of any ranking-reversal claim must be declared.

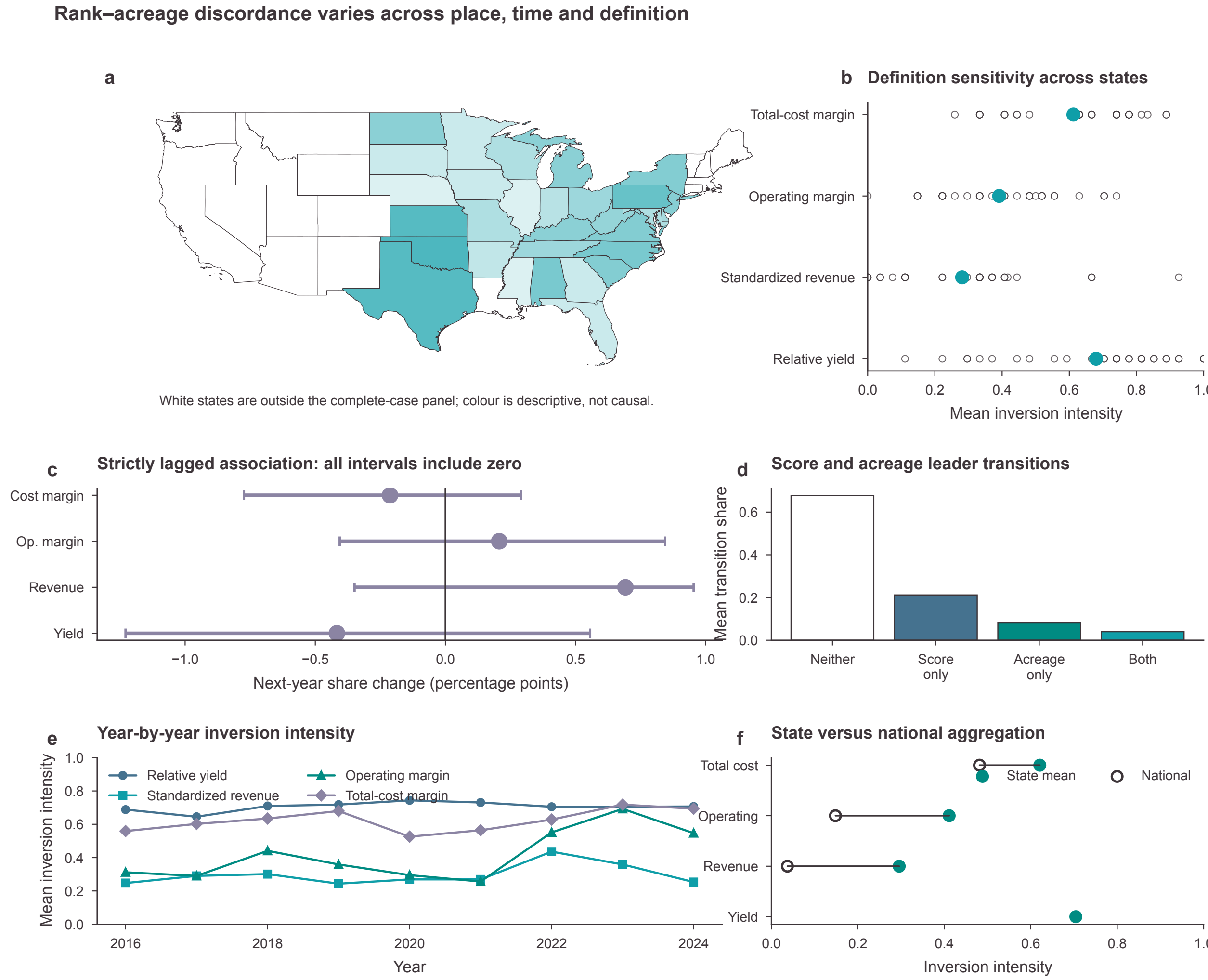

Figure 6: Spatial, definitional, temporal and aggregation boundaries in official data. The panel covers 248 complete state-years in 31 states from 2016 to 2024. The ranked state dot plot uses official state records; intervals use 5000 state-cluster draws. Every primary strictly lagged interval includes zero. Observed acreage is not interpreted as a model optimum.

Support checks reinforce that boundary. Omitting one state at a time leaves mean operating-margin inversion between 0.399 and 0.421. Annual coverage and missingness are shown explicitly, and no suppressed value is filled from another source. The map uses official Census state geometry only to locate admitted observations; white states are outside the complete-case panel. Supplementary Fig. S6 reports the ranked distributions, annual paths, lagged estimates, transitions and sample counts at larger scale.

The evidence boundary is asymmetric. Public data identify acreage, yield and transparent accounting relations, and they support descriptive rank agreement and transition estimands. They do not reveal private objectives, constraints, beliefs, downside-risk limits or a selected optimum. Model-linked signatures are therefore reported as compatibility checks, not mechanism tests. The full annual, leave-one-state-out and sample-flow checks appear in Supplementary Fig. S5.

6 Discussion

The central result is that ranking reversal is a relationship, not a single mechanism. It compares an ordinal statement with a cardinal allocation drawn from a constrained optimal face. Exact theory establishes why the ranking alone is insufficient, and controlled experiments identify two positive mechanism classes: operational interventions can displace the ranked leader, and the interaction between information and action-set flexibility changes sign with the actions that flexibility enables.

This mechanism family has a useful internal structure. Cardinal-margin reversal concerns the objective, operational displacement concerns the feasible set, downside-risk displacement concerns a tail-risk boundary, and set-valued reversal concerns the geometry of the optimum and its selector. The same Corn–Soybean acreage order can therefore correspond to different counterfactual predictions. A mechanism label is scientifically useful only when the corresponding object is observed, assigned or credibly estimated.

This distinction avoids two interpretive extremes. Treating every score–acreage mismatch as an error ignores objectives, constraints and uncertainty. Treating every mismatch as uninformative overlooks controlled settings in which the intervention and optimal face are known. The operational experiment supports its mechanism because the constraints are assigned and the entire optimal face is evaluated. The information experiment supports actionability because signal and action-set contrasts are factorial and exact checks hold. Neither experiment establishes population prevalence.

The inconclusive experiments are part of the result. E1, E3, E4 and E5 show that deterministic reproduction, solver agreement and visually structured contrasts do not guarantee the required precision. Preserving their intervals, stopping outcomes and infeasible rows prevents a passing sub-contrast from being misread as an experiment-level conclusion. Numerical integrity and inferential precision remain separate criteria.

The extended public-data panel establishes a different evidence layer. Concurrent rank–acreage inversion is widespread but definition- and state-dependent; all primary lagged intervals include zero; and national aggregation changes the pattern. These data do not measure a farm's feasible set or objective and cannot identify observed acreage as optimal. In particular, they support no inference about private downside-risk preferences, dependence mechanisms, welfare or regret relative to a latent optimum.

Several limitations remain. The empirical analysis covers three crops and uses national price and cost series to create transparent accounting scores, omitting local basis, heterogeneous costs and private expectations. The model is linear and single-period, excluding fixed charges, endogenous prices, spatial interaction and dynamic rotations. The simulation parameters are controlled design values rather than farm estimates. Future work requires decision-level observations on expectations, constraints and contracts before the controlled mechanisms can be connected to field prevalence.

The practical implication is procedural. A crop ranking intended to guide acreage should disclose the cardinal payoff mapping, operational constraints, joint uncertainty model, downside-risk convention and selection rule. When those objects are unavailable, score–acreage comparisons should remain descriptive. A prediction becomes prescriptive only through the decisions it can alter and the constraints under which those decisions are feasible.

This distinction matters beyond farm planning. A lender may use a crop ranking to screen a production plan, and a decision-support platform may display a recommended crop order. In either case, treating the top rank as a prescribed acreage share can penalize a producer for obeying a rotation, liquidity or contract restriction that the score omits. Conversely, describing every mismatch as rational would make the system unfalsifiable. The appropriate response is to expose the decision model and compare feasible counterfactuals, not to infer hidden constraints from disagreement alone.

Connecting the controlled mechanisms to field prevalence requires data at the decision unit. Useful measurements include pre-plant price and yield beliefs, available credit, crop-specific variable costs, land quality, rotation history, insurance, forward contracts, capacity bounds, signal receipt and the menu of actions actually considered. Repeated decisions would then allow researchers to distinguish persistent preferences from changing feasibility and information. Aggregate state acreage can motivate these questions, but it cannot answer them by itself.

7 Conclusion

An ordinal crop ranking does not identify a cardinal acreage allocation. Allocation follows only after cardinal margins, joint uncertainty, operational feasibility, tail-risk conventions and an optimizer-selection rule are supplied. Within that declared system, operational constraints can force universal reversal across the complete optimal face, and information can complement, substitute for or be unrelated to action-set flexibility.

The official 2016–2024 panel adds a descriptive boundary. Rank–acreage discordance varies across states, years, score definitions and aggregation, while every primary lagged interval includes zero. These observations make disagreement empirically visible but do not reveal private objectives, constraints or welfare.

Crop-ranking systems should therefore separate prediction from prescription. If a system recommends acreage, it should disclose the decision model that converts scores into feasible actions. If those objects are unavailable, the defensible claim is about ranking correspondence, not optimal land allocation.

8 Methods

8.1 Optimization model and optimal-face classification

Per-acre stochastic margins are combined linearly in acreage. Operational feasibility includes lower and upper crop bounds, land, budget and shared linear restrictions. Loss-CVaR is

$$r_\alpha(x) = \min_{v \in \mathbb{R}} \left\{ v + \frac{1}{1-\alpha} \mathbb{E}[(-\widetilde{\pi}^\top x - v)_+] \right\}, \tag{9}$$

with a free VaR auxiliary variable. In finite scenarios, non-negative excess-loss variables give a linear programme. Possible and universal reversal are computed by minimizing and maximizing pairwise acreage gaps over the objective-equivalent optimal face. Selected reversal uses the frozen deterministic selection rule. Direct loss-CVaR, primal feasibility, complementary slackness and stationarity are recomputed independently.

8.2 Confirmatory simulations

The E1–E6 design was specified before execution. Each experiment declares factors, estimands, multiplicity, sequential replication checks, precision targets and a ceiling. E2 crosses assigned budget, rotation and contract interventions against a fixed ranking and law and includes a dominant-crop-bound anchor. Its primary family contains allocation displacement, expected-margin and reversal-probability contrasts. E6 crosses three signal-accuracy levels with restricted and flexible action sets in three constructive environments. An experiment supports a main conclusion only when every primary interval meets its experiment-level target. Infeasible outcomes are retained and classified. Reverse-order replay and solver-method sensitivity rerun named tasks.

8.3 Mechanism attribution

The M0–M4 path adds cardinal margins, operations, downside risk and dependence to one closed benchmark. Because sequential contributions depend on order, every block coalition is solved and Shapley contributions are calculated for allocation, expected margin, loss-CVaR and concentration. The efficiency residual compares the contribution sum with the full-minus-base difference. KKT pressure terms are evaluated from Eq. 8 and interpreted as local accounting terms.

8.4 Official data and frozen empirical design

USDA NASS Crop Production annual summaries provide state and national planted acreage and yield for corn, soybeans and winter wheat. Non-overlapping reports cover 2016–2024. The parser's overlapping 2024 block reproduces the previously frozen table exactly. USDA ERS national harvest prices, operating costs and total costs are joined by crop-year; the registered BLS CPI-U series supplies the real-dollar conversion. A state-year is retained only when all three crops have acreage and yield. Missing and suppressed values are not imputed.

Four score definitions were frozen before extended results were inspected: relative yield, standardized revenue, operating margin and total-cost margin. Concurrent estimands are Kendall's tau-b, Spearman correlation when defined, pairwise inversion intensity and top-rank disagreement. The temporal model regresses acreage-share change on prior score-leader status and prior acreage share with crop, year and state fixed effects. Scores strictly precede the outcome. Uncertainty uses 5000 state-cluster bootstrap draws. All empirical estimands are descriptive.

Leader persistence records whether the top-scoring and top-acreage crops remain unchanged between adjacent years. Transition structure partitions each state-year transition into neither leader changing, only the score leader changing, only the acreage leader changing or both changing. Aggregation repeats the ranking calculations on national crop totals and preserves ties rather than imposing an arbitrary order. Leave-one-state-out analysis repeats the definition-level mean after omitting each admitted state. Missingness is summarized by year, crop and variable; support sensitivity is reported without imputation.

The official NASS and ERS series were audited forward, but the next year was excluded because a complete annual CPI-U value matching the frozen construction was unavailable at the design freeze. County reconstruction was not run because no targeted credential-free Quick Stats query was available and the public all-crops bulk snapshot was not a bounded extraction. Both decisions are recorded with source URLs, checksums and blockers; no county claim is made.

8.5 Visualization and reproducibility

The central colour card is `#3D3539`, `#0F9EA8`, `#008B82`, `#45728F`, `#8CD1B2` and `#8B84A3`. Corn, soybean and winter wheat are fixed to `#45728F`, `#008B82` and `#8CD1B2`; supported evidence uses `#0F9EA8`, unresolved evidence uses `#8B84A3`, and text and axes use `#3D3539`. Each figure is exported as editable SVG/PDF, 300-dpi PNG and 600-dpi TIFF and inspected at its 183-mm design size in full colour, grayscale, deuteranopia and protanopia simulations.

For every main scientific question, two structurally different concepts were rendered at 183 mm before the final design was selected or combined. Final layouts reserve separate lanes for titles, labels and graphical marks. Renderer-bound checks reject clipping and title collisions, and full-size inspection verifies that annotations do not cover data marks. The colour validator permits white and transparent derivatives of the registered card but rejects unregistered base hues. Crop and definition encodings use hatch, marker shape, line style or position in addition to colour. Official 2024 Census generalized state geometry supplies only the map scaffold.

Raw sources, tidy outputs, source tables, figures and manuscript numbers carry SHA-256 lineage. The empirical pipeline is accepted only when two isolated runs are byte-identical. The manuscript and Supplementary Information are compiled in isolated directories; citations, references, fonts, page renders and archive checksums are checked before release.

Declarations

Acknowledgements. To be completed by the authors before submission.

Author contributions. To be completed using the journal's CRediT format.

Competing interests. The authors must confirm the final statement before submission.

Ethics. The study uses public aggregate data and no human-participant or animal data.

Data availability. All admitted source identifiers, immutable raw-file checksums, processing contracts and derived analysis files are recorded in the repository's data and evidence registries. Public-source redistribution remains subject to the originating agencies' terms.

Code availability. The complete theory, optimization, simulation, empirical, visualization and manuscript pipelines are versioned in the companion repository. Reproduction commands and checksums are provided with each artifact class.

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